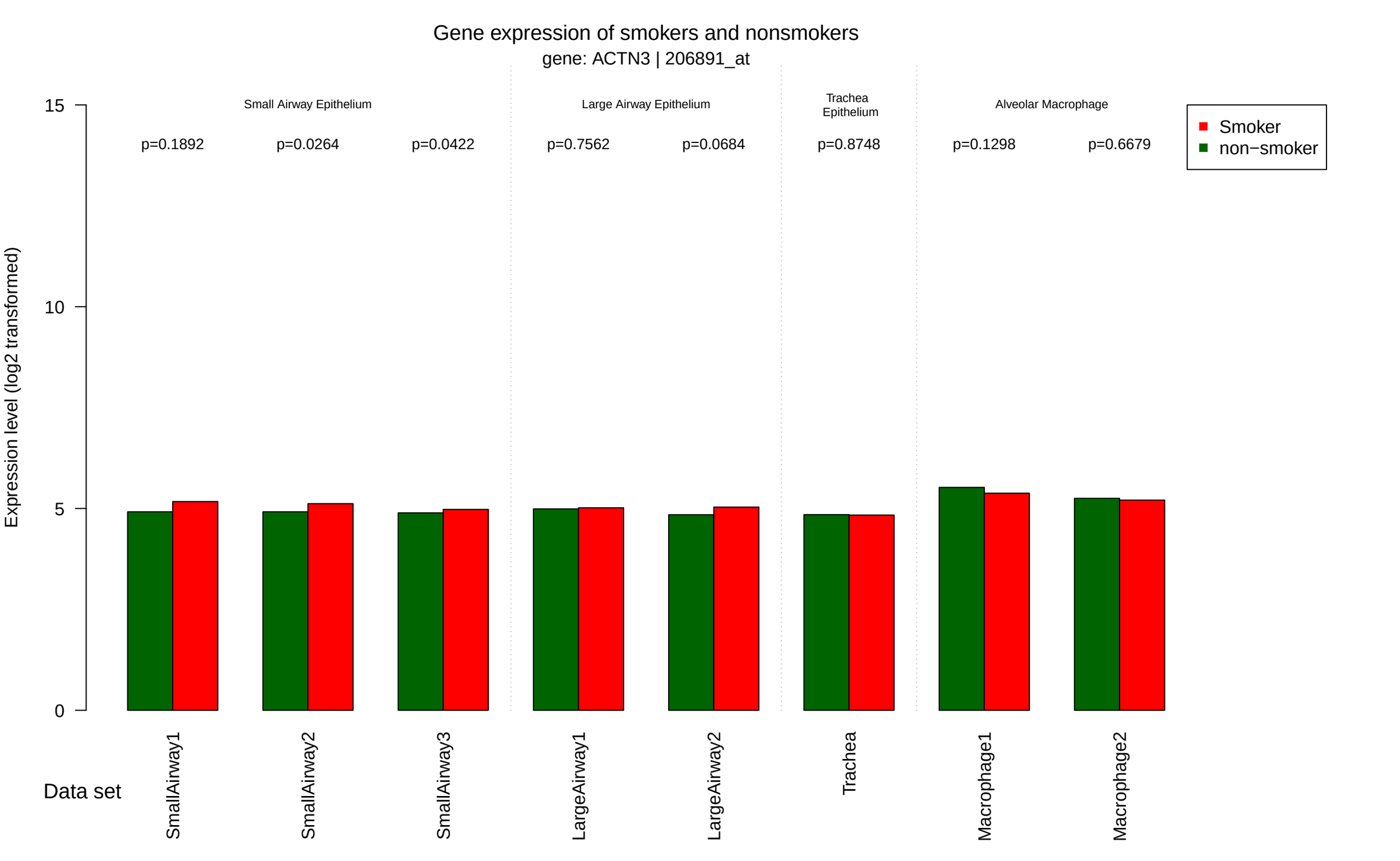
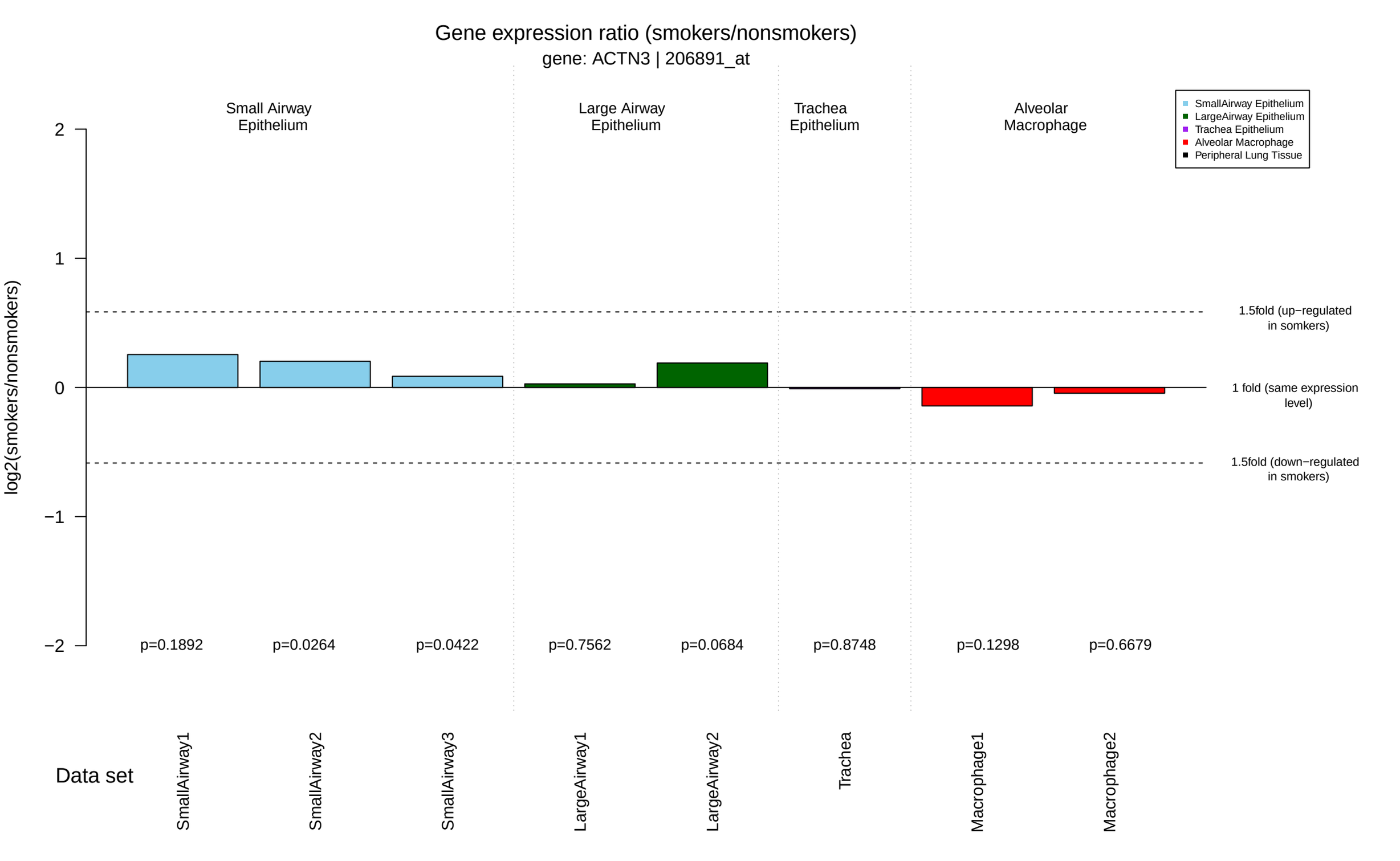
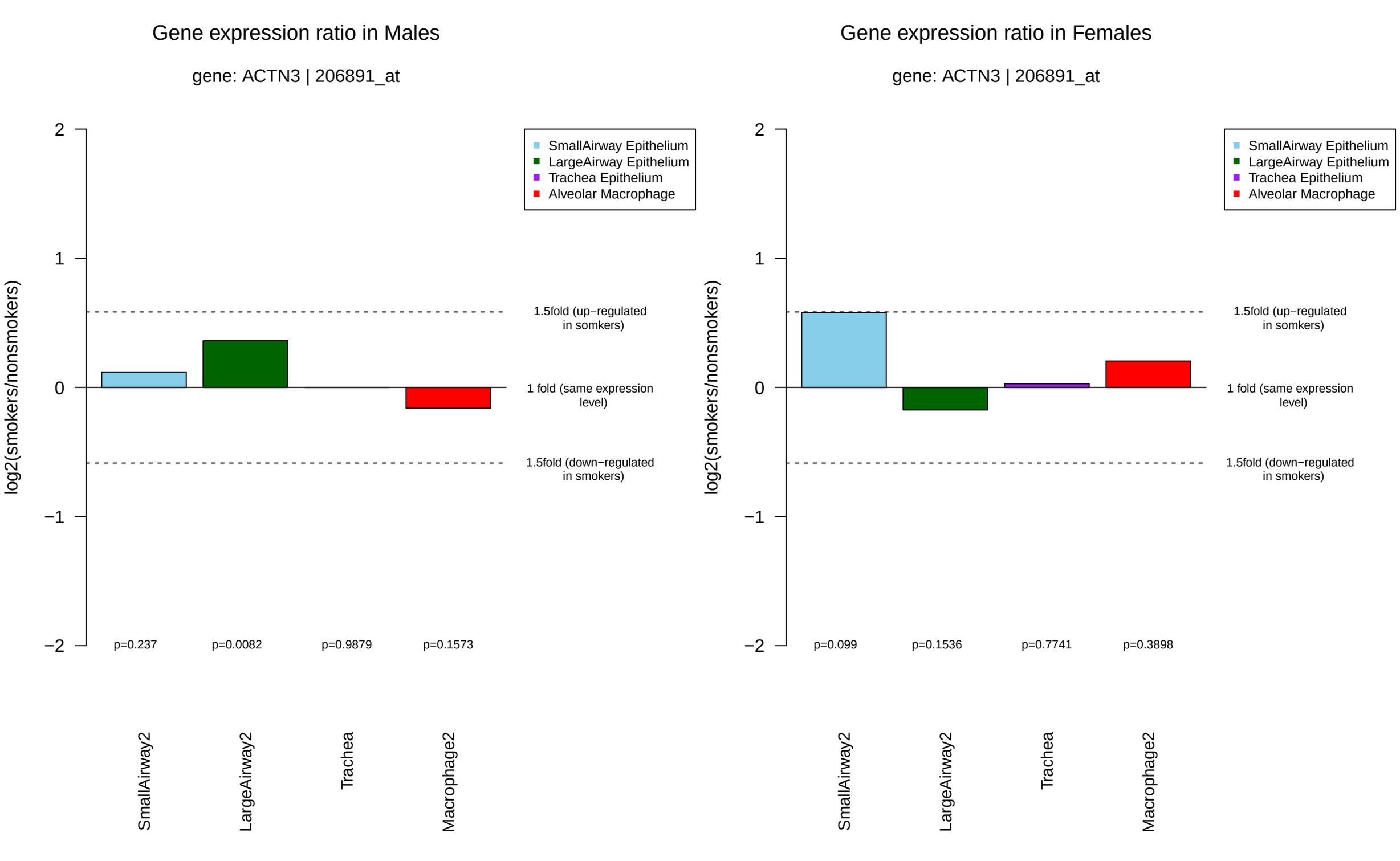
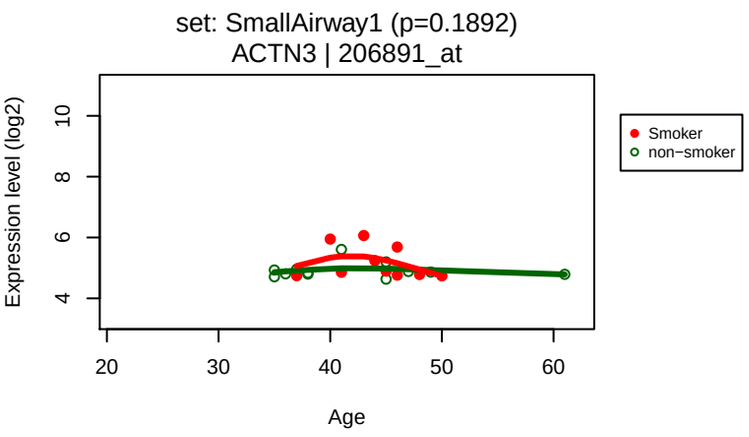
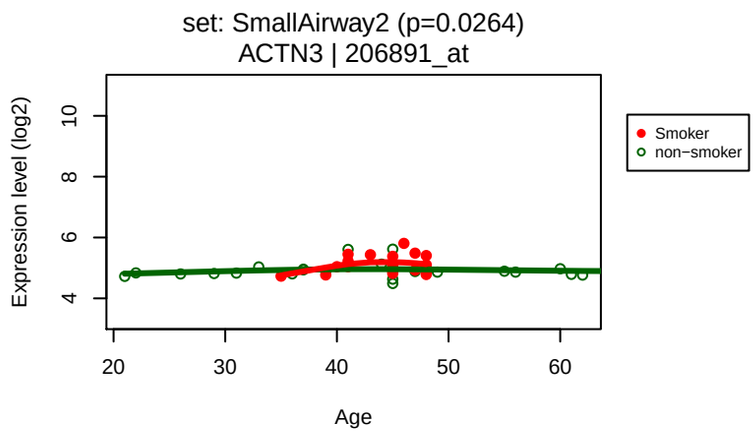
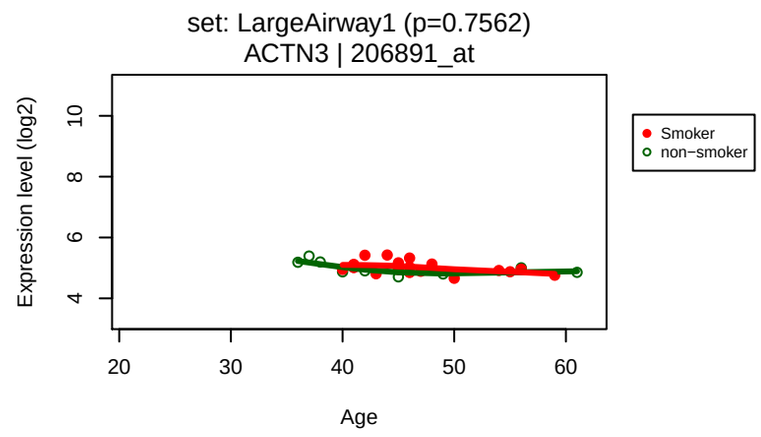

Gene expression of smokers and nonsmokers
gene: ACTN3 | 206891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1892
p=0.0264
p=0.0422
p=0.7562
p=0.0684
p=0.8748
p=0.1298
p=0.6679
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

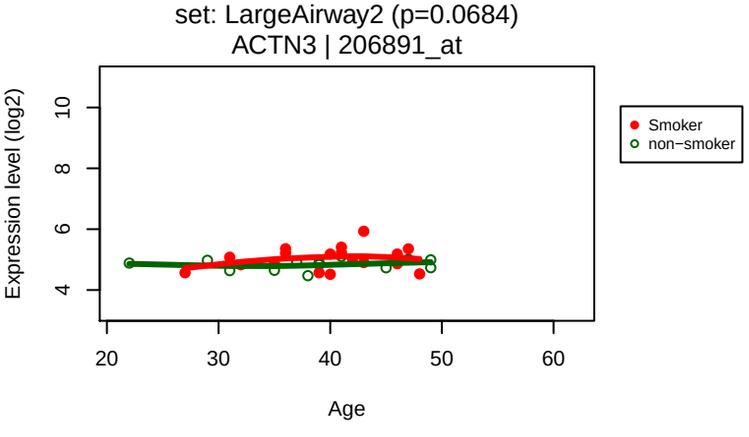

Gene expression ratio (smokers/nonsmokers)
gene: ACTN3 | 206891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1892
p=0.0264
p=0.0422
p=0.7562
p=0.0684
p=0.8748
p=0.1298
p=0.6679
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

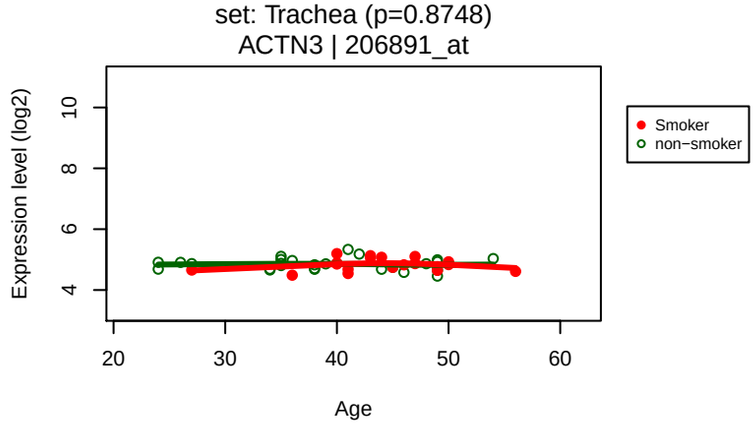

Gene expression ratio in Males
gene: ACTN3 | 206891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.237
p=0.0082
p=0.9879
p=0.1573
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTN3 | 206891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.099
p=0.1536
p=0.7741
p=0.3898
SmallAirway2
LargeAirway2
Trachea
Macrophage2

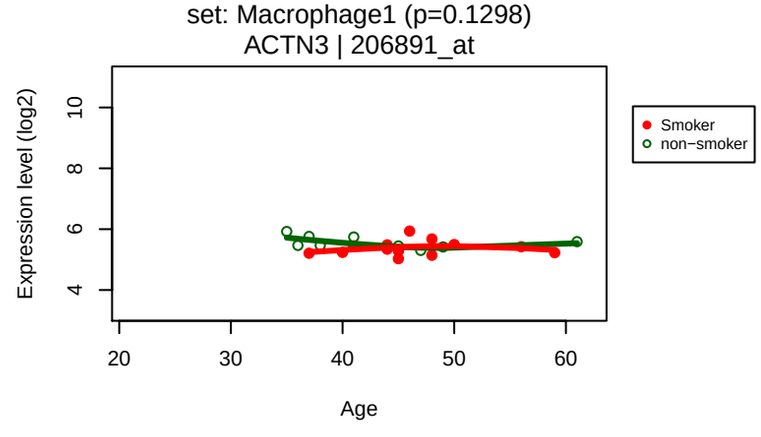

set: SmallAirway1 (p=0.1892)
ACTN3 | 206891_at
Expression level (log2)
Age
Smoker
non−smoker

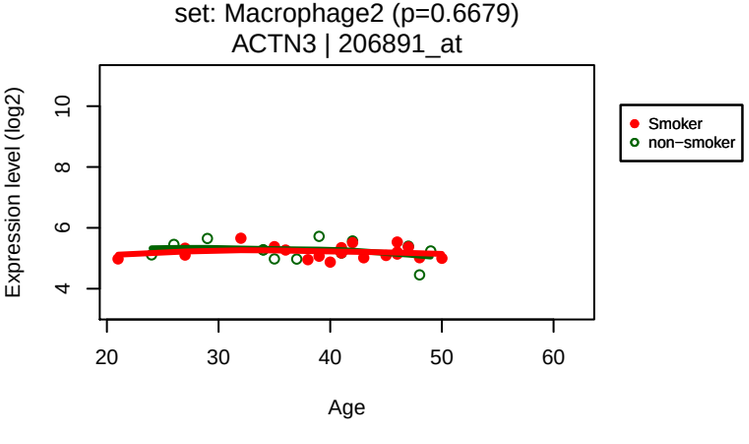

set: SmallAirway2 (p=0.0264)
ACTN3 | 206891_at
Expression level (log2)
Age
Smoker
non−smoker

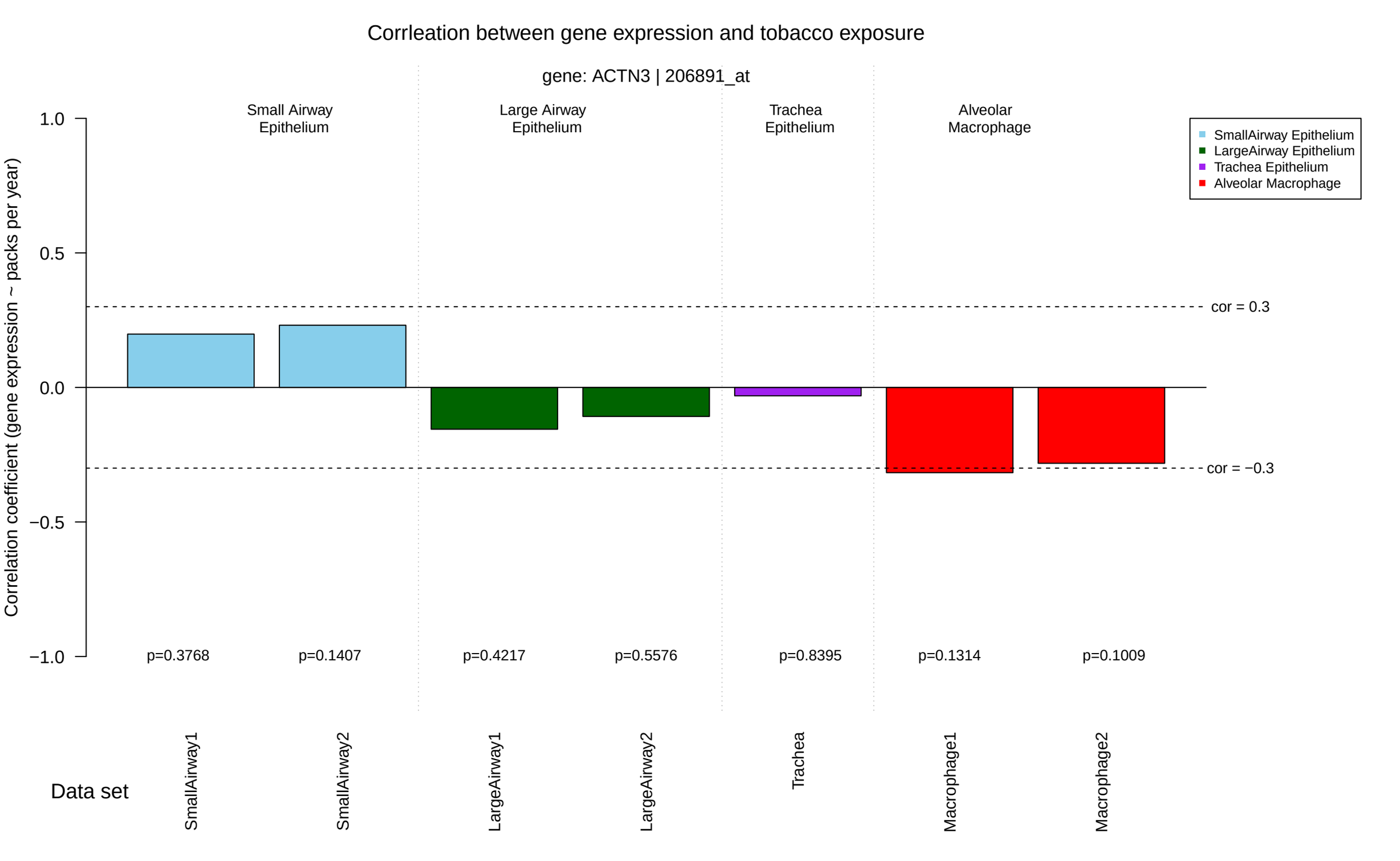

set: LargeAirway1 (p=0.7562)
ACTN3 | 206891_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0684)
ACTN3 | 206891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8748)
ACTN3 | 206891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1298)
ACTN3 | 206891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6679)
ACTN3 | 206891_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTN3 | 206891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3768
p=0.1407
p=0.4217
p=0.5576
p=0.8395
p=0.1314
p=0.1009
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2